Book Price and Rating Analysis using Web Scraping and Machine Learning

Step 1 : Web Scraping Script:

- The dataset was collected using web scraping techniques from the publicly available website Books to Scrape.
- Python libraries such as Requests and BeautifulSoup were used to automatically extract book information from multiple pages of the website.
- The script iterated through 50 pages and collected details including book title, price, rating, and availability.
- A total of 1000 book records were collected and stored in a CSV file named books_raw.csv for further **analysis.**

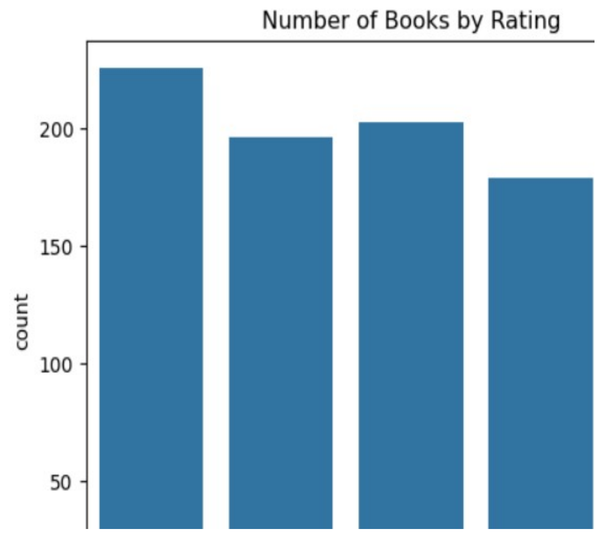
Step 2 : Data Cleaning & Preprocessing :

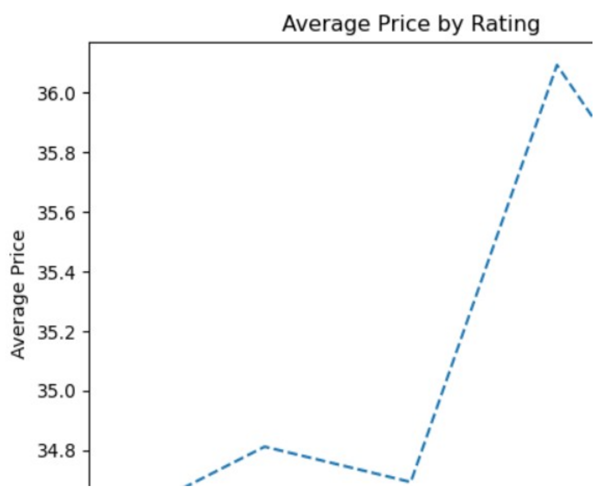
- The raw dataset obtained through web scraping required preprocessing before analysis. Missing values and duplicate records were checked and removed to ensure data quality.
- The price column contained currency symbols which were removed and converted into numeric format. The rating column originally contained text values such as "One", "Two", etc., which were mapped to numeric values from 1 to 5.

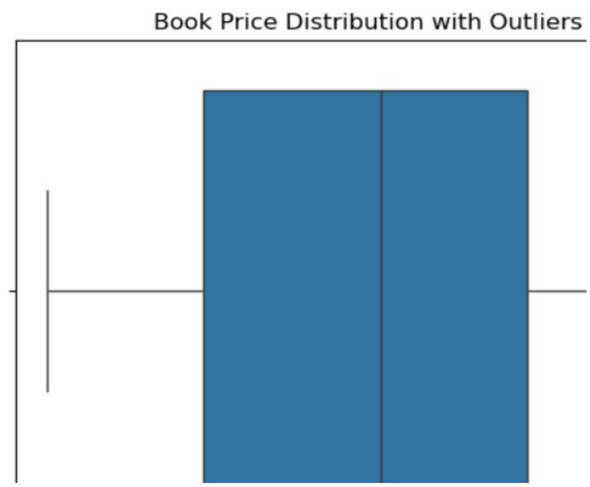
Step 3 : Exploratory Data Analysis & Visualization:

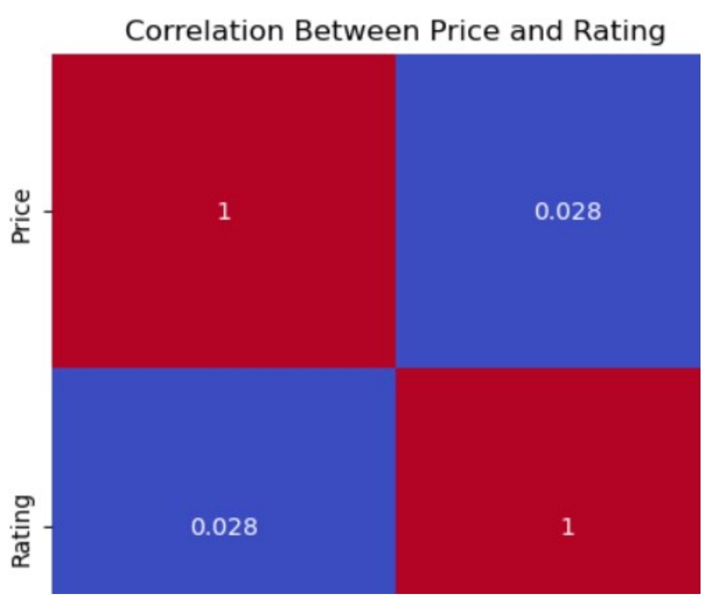
- Exploratory Data Analysis was conducted to understand the patterns and relationships within the dataset. Summary statistics such as mean, median, minimum, and maximum values were calculated to analyze the distribution of book prices.
- Several visualizations were created including bar charts, scatter plots, boxplots, and heatmaps. The rating distribution showed that most books have ratings between **3 and 4 stars**. The price distribution indicated that the majority of books fall within the **£20 to £40 range**.
- The scatter plot between price and rating revealed that there is **no strong relationship between book price and rating**. Additionally, we used Box Plot for Outliers but we don’t find any outliers .
- Overall, the analysis provided useful insights into pricing patterns and rating distribution of books in the dataset.


Price vs Rating Trend
Price
60
50
40
30
20


Number of Books by Rating
count
200
150
100
50


Average Price by Rating
Average Price
36.0
35.8
35.6
35.4
35.2
35.0
34.8


Book Price Distribution with Outliers


Correlation Between Price and Rating
Price
Rating
1
0.028
0.028
1

Step 4 : Model Building & Evaluation :

- A Linear Regression model was implemented to predict book prices using ratings as the input feature. The dataset was split into training and testing sets using an 80–20 ratio. The model was evaluated using Mean Squared Error (MSE) and R^2 score.
- The results showed an MSE of approximately 211 and an R^2 score close to zero, indicating that book rating has little influence on book price. This observation is consistent with the findings from the exploratory data analysis.
- A Decision Tree Regressor was implemented to predict book prices using ratings as the input feature. The model was trained on the training dataset and evaluated using Mean Squared Error (MSE) and R^2 score.
- The results showed an MSE of approximately 212 and a negative R^2 score, indicating that rating alone is not a strong predictor of book price. The performance was similar to the Linear Regression model.
- A Random Forest Regressor was implemented as the third machine learning model to predict book prices. Random Forest is an ensemble learning method that combines multiple decision trees to improve prediction accuracy. The model was trained using the training dataset and evaluated using Mean Squared Error (MSE) and R^2 score. The results were compared with Linear Regression and Decision Tree models to identify the best-performing approach.
- In this phase, three regression models including **Linear Regression, Decision Tree Regressor, and Random Forest Regressor were implemented to predict book prices using rating as the input feature.** The dataset was split into training and testing sets using an 80–20 ratio.
- The models were evaluated using Mean Squared Error (MSE) and R^2 score. The results indicated that all models showed weak predictive performance, suggesting that **book rating alone is not a strong predictor of price**. These findings are consistent with the insights obtained from the exploratory data analysis.

Step 5 : Interactive Dashboard :

- An interactive dashboard was developed using Power BI to visualize key insights from the dataset. The dashboard includes key performance indicators such as total number of books and average book price.
- Several visualizations including rating distribution charts, price band analysis, and scatter plots were used to identify trends and patterns in the data. AI visuals such as Key Influencers were used to analyze factors influencing book prices.

Total Books

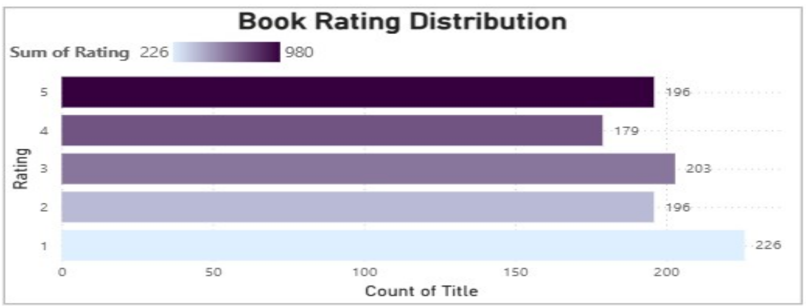
1000

Count of Title

Average Book Price

35.07

Average of Price



Insights :

The exploratory analysis revealed several useful insights. Most books in the dataset have ratings between **3 and 4 stars**, and the majority of book prices fall within the **£20 to £40 range**. The scatter plot analysis showed that there is **no strong relationship between book rating and price**, indicating that book pricing may depend on other factors not present in the dataset.

Model Results :

Three machine learning models were implemented to predict book prices: **Linear Regression, Decision Tree Regressor, and Random Forest Regressor**. The models were evaluated using **Mean Squared Error (MSE)** and **R^2 score**. The results showed that all models had relatively low predictive performance, confirming that rating alone is not a strong predictor of book price.

Recommendations :

Based on the findings, it is recommended that additional variables such as **book category, author popularity, number of reviews, and publication details** be included in future analysis to improve predictive performance. Expanding the dataset with more relevant features would allow more accurate modeling and deeper insights into pricing patterns.